Cybercrime: Challenges in Law Enforcement's Ability to Keep Up with Rapid Technological Advancements

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Abstract

This paper examines the challenges modern law enforcement faces in keeping up with rapidly evolving technology and cybersecurity threats. One of the threats that has emerged from developing technology is the threat of ransomware. Ransomware poses a significant threat to both our national infrastructure and security, with the potential to shut down systems ranging from small businesses to entire cities. The challenge becomes even more complex with these attacks are carried out by international actors. In response to this, the U.S. Federal Bureau of Investigation (FBI) collaborates closely with international partners to identify and disrupt ransomware networks. These partnerships enable the FBI to make arrests, takedown infrastructure, and disrupt the source of income for cybercriminals. As ransomware threats continue to advance, continued cooperation between the FBI and domestic and international partners is essential in maintaining U.S. national security.

Keywords: ransomware, cybersecurity, cybercrime, Federal Bureau of Investigation, FBI, law enforcement, technology

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As technology constantly evolves, law enforcement faces an ongoing challenge to keep up with the growth of cybercrime. One of these ongoing challenges is the issue of ransomware. Ransomware is a malicious program created and designed to hold a computer system hostage until a ransom is paid (Barker et al., 2022). According to Vorndran (2021), the most common targets for ransomware attacks are large businesses and municipalities. An aspect of ransomware that makes it extra challenging for law enforcement is that international attackers (Vorndran, 2021) often perpetrate it. Our nation's ability to defend against such attacks often relies heavily on the cooperation with the nation that the attack originated from (Vorndran, 2021). One of the law enforcement agencies primarily involved in leading international ransomware attack investigations is the Federal Bureau of Investigation (FBI). For the FBI, they see the main solution in combatting ransomware is maintaining "strong foreign partnerships" and increasing its partnership with the private sector (Vorndran, 2021). Because ransomware attacks often originate internationally, the FBI must rely on cooperation from domestic and international partnerships to effectively combat these attacks.

Ransomware is a program created by bad actors for the purpose of shutting down a computer, computer systems, or entire servers (Vorndran, 2021). The program typically accomplishes this by first infiltrating the system and then encrypting the data and contents of the system (Vorndran, 2021). This, as Vorndran (2021) explains, disables the system entirely. Once a system has been successfully held captive by the ransomware, the victim of the attack cannot gain access to their system until a "ransom" has been paid (Barker et al., 2022). A ransomware attack is like a group of criminals breaking into your home, changing all the locks, and then

demanding payment in exchange for the new keys. Ransomware stands in contrast to other forms of cyberattacks in that it is not after data or trade secrets, but rather cash through extortion, typically in the form of untraceable cryptocurrency (Vorndran, 2021).

As Vorndran (2021) explains, because money is the primary goal of a ransomware attack, the victims are usually businesses and municipalities. These are common targets because of the urgency typically involved in restoring their business operations or city services (Vorndran, 2021). This is what creates some of the key issues of ransomware attacks. First, ransomware attacks create large financial burdens on small to midsize businesses, not only because of the ransom itself but also the cost of not being able to operate (Vorndran, 2021). Second, ransomware attacks can shutdown businesses involved in vital infrastructure such as healthcare and transportation (Vorndran, 2021). Lastly, ransomware attacks can effectively shut down a city government, and in some cases as Vorndran (2021) states, even disrupt operations at the state level. An example of this happened in 2018 when the city of Atlanta experienced major service outages due to a ransomware attack by two Iranian hackers (Vorndran, 2021). As Vorndran (2021) points out in this case, one of the key complications for law enforcement is that these attacks are often conducted by international hacking groups, making investigation and prosecution much more difficult for law enforcement.

International ransomware attacks are not by nature always difficult to deal with. In a number of cases, United States law enforcement is able to effectively partner with other nations and international agencies to prosecute these attackers (Vorndran, 2021). The trouble primarily arises from attacks originating in countries where the US has no jurisdiction or partnerships (Vorndran, 2021). In those cases, Vorndran (2021) explains, US law enforcement simply has to hope that those countries will cooperate when asked. However, this gets especially complicated

when actors that initiated the attack are based within countries openly hostile to the US such as Russia, Iran, China, and North Korea (Vorndran, 2021). Because of the safe havens that these countries provide to ransomware groups, these groups are able to operate with little hinderance (Vorndran, 2021).

In combination to the challenge that distance poses, US law enforcement also faces the difficulty of a digital infrastructure that enables ransomware attacks and makes the perpetrators hard to track down (Vorndran, 2021). For example, some developers do not carry out the attacks themselves, instead, they offer ransomware as a service (RaaS) to other criminal groups, further obfuscating who the actual perpetrator of the crime is (Vorndran, 2021). In addition to that, attackers often use VPN services to mask their true location, making it difficult for law enforcement to track down the origin of an attack (Vorndran, 2021). As Vorndran (2021) explains, the attackers will often use VPN services that are willing to ignore their activities. Lastly, ransomware attackers tend to only take payment in the form of difficult to trace cryptocurrency (Barker et al., 2022). As Vorndran (2021) points out, the unregulated and fluid nature of cryptocurrency makes it a common choice for attackers, since it can be easily transferred globally with little trace. All of these factors together create a scenario in which finding cybercriminals can be impossible. However, despite this, agencies like the Federal Bureau of Investigation (FBI) remain committed to combatting cybercrimes like ransomware.

There are three key ways that the FBI combats Ransomware. First, by targeting the individuals and groups perpetrating the attacks directly (Vorndran, 2021). Second, by dismantling the infrastructure that allows these attacks to happen (Vorndran, 2021). Third, by undercutting ransomware groups' sources of income (Vorndran, 2021). The FBI combats individual perpetrators through a network of partnering nations and agencies (Vorndran 2021).

Vorndran (2021) states that in 2021, the FBI was able to successfully arrest a Ukrainian citizen guilty of multiple ransomware attacks on a Miami based IT company “Kaseya”. Second, the FBI dismantles the infrastructure that allows for ransomware attacks in order to prevent further attacks, increase the operation costs of ransomware groups, and gain intelligence on attacker methods (Vorndran, 2021). Vorndran (2021) recounts that in June of 2021, through the aid of international partners and agencies, were able to shut down the service “DoubleVPN” that was complicit in aiding in and providing services to ransomware groups. Lastly, Vorndran (2021) explains that the FBI attacks ransomware by finding ways to trace cryptocurrency exchanges and restore ransoms back to victims. In July of 2021, the FBI in conjunction with the Office of Foreign Assets Control was able to Sanction “Chatex Virtual Currency” for being complicit in enabling currency transfers related to ransomware attackers (Vorndran, 2021).

For the future, the FBI proposes two solutions to continue to fight the global threat of ransomware. First, to maintain and expand domestic and international partnerships. Second, to increase security resources, infrastructure, frameworks and requirements for the United States private sector (Vorndran, 2021). Without the aid of its’ network of partners, it would be impossible for the FBI to effectively combat ransomware (Vorndran, 2021). As Vorndran (2021) explains, the FBI relies on expanding its team of partners to reliably detect, locate, and arrest cybercriminals internationally. It is through these increased partnerships that allows the FBI to increase its body of information that enables it to inform victims, provide incite to legislation, and strengthen national cyber defenses (Vorndran, 2021).

Since most of the United States’ infrastructure is own by the private sector, the FBI has a vested interest in not only partnering with these businesses, but also providing them with security resources, frameworks, and standards (Barker et al., 2022). The FBI doe this first by providing

up to date threat information regarding new threat groups and emerging ransomware attack methods (Vorndran, 2021). Second, in conjunction with other federal agencies like the National Institute of Standards of Technology (NIST) and the Cybersecurity and Infrastructure Security Agency (CISA), the FBI provides security resources, frameworks, and institutional requirements for businesses (Cybersecurity and Infrastructure Security Agency, 2020). Third, the FBI works to make it as easy as possible for organizations and businesses to report ransomware and other cybersecurity attacks (Vorndran, 2021). As Vorndran (2021) explains, the FBI also advocates for legislation that would place greater requirements for ransomware reporting. This is because the FBI views timely reporting as an essential for staying informed about international cyber threats and protecting national security (Vorndran, 2021). According to the FBI, these two combined efforts provide a path forward in defending our nation against future ransomware attacks (Vorndran, 2021).

Because of the ongoing international threat that is ransomware, the U.S. will continue to require support from its internal agencies like the FBI, as well as its domestic and international partnerships. The threat of ransomware continues to burden our nation's private and civil infrastructure (Vorndran, 2021). The international nature of ransomware attacks requires that law enforcement rely on global partners and agencies to combat attackers (Vorndran, 2021). The FBI leads this charge by targeting individual ransomware attackers and groups, dismantling infrastructure that enables ransomware, and cutting off avenues of income streams that support these attackers (Vorndran, 2021). To continue these efforts, the FBI requires the aid of the U.S. private sector to harden their cybersecurity defenses through recommended practices (Cybersecurity & Infrastructure Security Agency, 2020). Without law enforcement agencies such as the FBI, our nation and its citizens would be subject to constant cyber-attacks.

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